# Wafer Map Tool

## Technical Documentation

Architecture · Pipeline · API · UI Reference

Version 2.3

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# 1. System Overview

The **Wafer Map Tool** is a desktop application for analysing semiconductor wafer measurement data. It provides a PyQt6 graphical interface and a headless programmatic API. Given a plain-text measurement file it produces three output artefacts: a multi-page PDF report, a colour-summary CSV, and an optional PowerPoint presentation.

## 1.1 Technology Stack

| Layer | Library / Framework | Purpose |
|---|---|---|
| Frontend UI | PyQt6 | Main window, widgets, threading |
| Visualisation | Matplotlib (Agg backend) | Wafer contour maps, charts, PDF pages |
| Data Processing | NumPy · Pandas · SciPy | Array maths, DataFrames, griddata interpolation |
| PDF Export | Matplotlib PdfPages + pypdf | Multi-page PDF + bookmarks |
| PPTX Export | python-pptx (optional) | PowerPoint slides |
| Logging | Python logging | File + console, session separators |
| Threading | QThread + threading.Event | Non-blocking background worker |

## 1.2 Entry Point

**main.py** is the application launcher. It:

- Installs a global sys.excepthook so unhandled exceptions are written to the log file before the process exits.
- Calls session_start() to write a timestamped separator to **wafer_tool.log**.
- Creates a QApplication, instantiates **DataProcessorUI**, and enters the Qt event loop.
- The same package can be imported headlessly (from wafer_tool import PlotConfig, generate_report) without touching Qt.

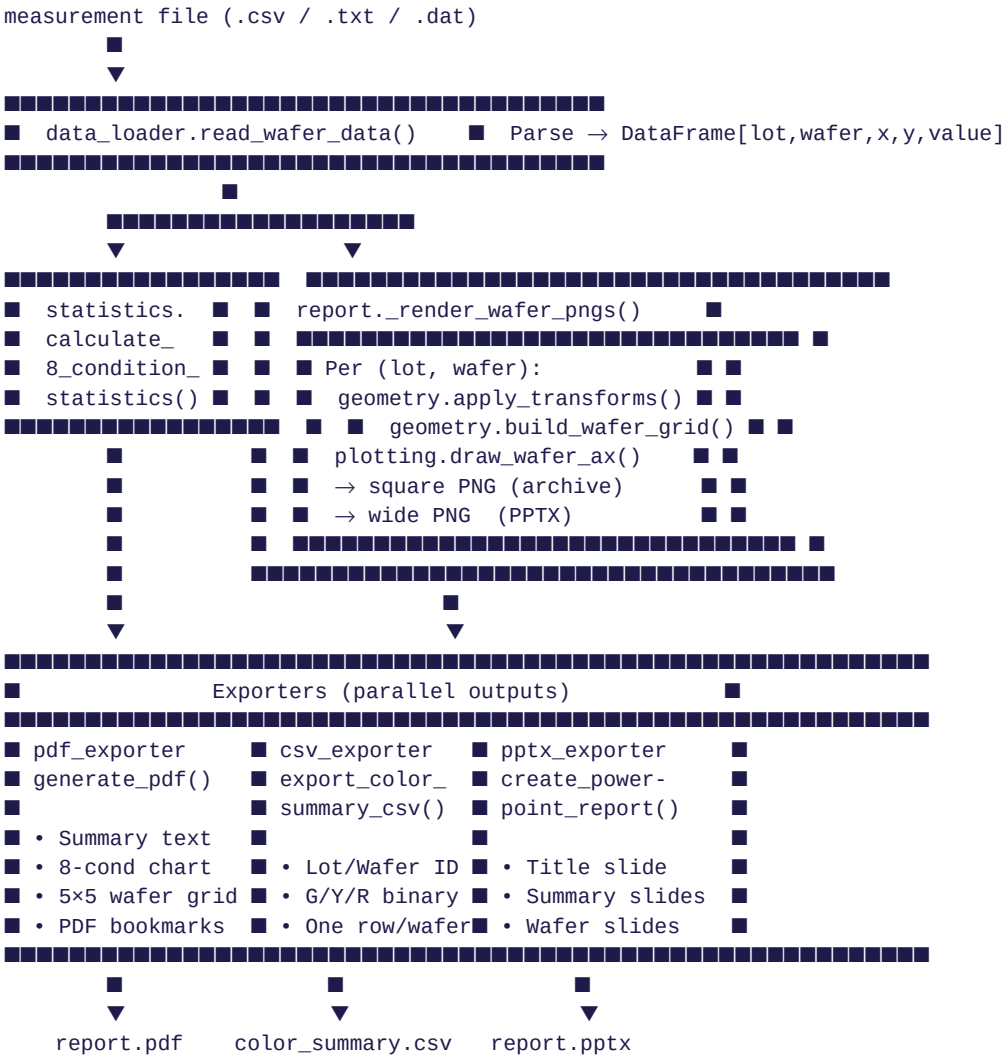
## 1.3 File Structure

```
E:\claude\Wafer_tool_NEW2\
■■■ main.py                          Entry point — PyQt6 GUI launcher
■■■ generate_docs.py                 This documentation generator
■■■ wafer_tool.log                   Append-only session log
■■■ wafer_tool\
    ■■■ __init__.py                  Public API surface
    ■■■ config.py                    PlotConfig dataclass + constants
    ■■■ data_loader.py               CSV / whitespace file parser
    ■■■ report.py                    Pipeline orchestrator
    ■■■ statistics.py                8-condition classification & bar chart
    ■■■ geometry.py                  Pure maths: transforms, interpolation
    ■■■ plotting.py                  draw_wafer_ax() contour renderer
    ■■■ logger.py                    Centralised logging setup
    ■■■ exceptions.py                Custom exception hierarchy
    ■■■ worker.py                    ReportWorker (QThread)
    ■■■ exporters\
    ■   ■■■ pdf_exporter.py          Multi-page PDF with bookmarks
    ■   ■■■ csv_exporter.py          Per-wafer colour-summary CSV
    ■   ■■■ pptx_exporter.py        PowerPoint report (optional)
    ■■■ ui\
        ■■■ main_window.py           DataProcessorUI + all inline widgets
        ■■■ histogram_widget.py      Integrated histogram + stats bar
        ■■■ orientation_widget.py    Reusable wafer-orientation selector
```

```
■■■ threshold_viz_widget.py  Threshold band visualisation
```

# 2. Processing Pipeline

The pipeline is orchestrated by **report.py** → **generate_report()**. All heavy computation runs on a background **QThread** (worker.py) so the UI remains responsive. The diagram below shows the complete data flow.




## 2.1 Step-by-Step Walkthrough

### Step 1 — Load Data

Called via read_wafer_data(filepath). Tries two parse strategies in order:

- CSV (comma-separated) — fast C engine.
- Whitespace/tab-separated — Python engine (required for regex separator \s+).

Accepts two input layouts:

- **5-column**: Lot | Wafer | X | Y | Value
- **4-column**: "LotID WaferID" (space-joined) | X | Y | Value

All numeric columns are coerced to **float32** to halve memory. Lot and wafer IDs use **category** dtype. Rows with NaN are dropped.

### Step 2 — Pre-compute Statistics

Called via calculate_8_condition_statistics(data, config). Classifies each wafer with ≥4 die points into one of **8 conditions** based on which colour zones are present:

| Code | Meaning | Zones Present |
|---|---|---|
| 100 | Green only | High zone only |
| 110 | Green + Yellow | High + Mid zones |
| 010 | Yellow only | Mid zone only |
| 111 | Green + Yellow + Red | All three zones |
| 101 | Green + Red | High + Low zones |
| 011 | Yellow + Red | Mid + Low zones |
| 001 | Red only | Low zone only |
| 000 | No colour | No die assigned (< 4 pts or all in one zone edge) |

Zone boundaries are derived from PlotConfig.t_low and t_high. Colour polarity is controlled by high_is_green:

- **high_is_green=True**: value > t_high → Green, t_low–t_high → Yellow, < t_low → Red.
- **high_is_green=False**: colour assignment reversed.

## Step 3 — Render Wafer PNGs

The inner loop in _render_wafer_pngs() runs once per (lot, wafer) pair:

- Creates a square 8×8 inch, 100 dpi Matplotlib figure.
- Calls draw_wafer_ax() which applies transforms, builds the 2-D interpolation grid, and renders a contourf map.
- Saves to individual_pngs_{timestamp}/.
- If python-pptx is installed, creates a second wide 8×6 inch figure with legend to temp_pptx_images/.
- Immediately closes and nulls each figure before opening the next to keep peak memory to one figure at a time.
- Fires on_wafer_ready(lot_id, wafer_id, png_path) callback for the UI live-preview grid.

## Step 4 — PDF Export

Calls generate_pdf(). The PDF is built with Matplotlib's **PdfPages** context and then post-processed with **pypdf** to inject named bookmarks. Structure:

- Summary text page(s): file path, timestamp, lot table (55 lines per page).
- 8-Condition bar chart page (from statistics module).
- Per-lot wafer grid pages: 5×5 = 25 wafers per page, each with its contour map, legend, and page header.
- PDF bookmarks: Summary Report → 8-Condition Distribution → one per lot.

## Step 5 — CSV Export

Calls export_color_summary_csv(). Re-reads the data file and for each wafer with ≥4 points emits a row:

- Columns: **Lot ID | Wafer ID | Green | Yellow | Red** (binary 0/1).
- Returns None gracefully if no valid wafers or file cannot be re-read.

## Step 6 — PowerPoint Export

Calls create_powerpoint_report() if python-pptx is installed.

- Searches for **Infineon Default Theme.pptx** template; falls back to python-pptx blank presentation.
- Slides: Title → Summary slides → One slide per wafer PNG (sorted alphabetically).

• Temp directories (temp_pptx_images/, temp_summary_images/) are deleted after the PPTX is written.

# 3. Module Reference

## 3.1 config.py — Configuration & Constants

### PlotConfig (dataclass)

| Field | Type | Default | Description |
|---|---|---|---|
| t_low | float | — | Lower threshold boundary (auto-sorted with t_high) |
| t_high | float | — | Upper threshold boundary |
| use_log | bool | False | Apply log10 scale to measurement values |
| high_is_green | bool | False | Color polarity: True → high values are green |
| mirror_x | bool | False | Flip wafer map horizontally |
| mirror_y | bool | False | Flip wafer map vertically |
| rot_deg | int | 0 | Clockwise rotation: 0, 90, 180, or 270 degrees |

**Derived properties:**

- color_scheme → **ColorScheme** with low/mid/high hex colours.
- log_suffix → empty string or "_LOG" (used in filenames).

### ColorScheme (frozen dataclass)

Holds three hex colour strings: low, mid, high.

- **Factory:** ColorScheme.from_mode(high_is_green: bool)
- high_is_green=True → low=#CC0000 (red), mid=#FFD700 (yellow), high=#00CC00 (green)
- high_is_green=False → low=#00CC00 (green), mid=#FFD700 (yellow), high=#CC0000 (red)

### Module-Level Constants

| Constant | Value | Purpose |
|---|---|---|
| GRID_RESOLUTION | 100 | Interpolation points per axis (100×100 grid) |
| WAFER_RADIUS_FACTOR | 1.05 | Expand boundary 5 % beyond data extent |
| MIN_POINTS_FOR_PLOT | 4 | Skip wafers with fewer die points than this |
| MAX_WAFERS_PER_PAGE | 25 | 5×5 wafer thumbnails per PDF page |
| LINES_PER_SUMMARY_PAGE | 55 | Text lines per summary page in PDF |

## 3.2 data_loader.py — File Parsing

### read_wafer_data(filepath, ...) → pd.DataFrame

Reads and validates a wafer measurement file. Accepts comma-separated or whitespace-delimited text. Column format:

| Layout | Col 0 | Col 1 | Col 2 | Col 3 | Col 4 |
|---|---|---|---|---|---|
| 5-column | Lot ID | Wafer ID | X | Y | Value |
| 4-column | "Lot Wafer" | X | Y | Value | (none) |

**Output DataFrame columns:** lot (category), wafer (category), x (float32), y (float32), value (float32). NaN rows are dropped.

**Errors:** raises DataLoadError on file not found, too few columns, or no valid rows after cleaning.

### Internal: _try_read(filepath) → pd.DataFrame

- Strategy 1: comma separator, C engine (fastest).
- Strategy 2: regex \s+, Python engine (required for regex).
- Raises DataLoadError if both strategies fail or neither produces ≥4 columns.

## 3.3 geometry.py — Coordinate Transforms & Interpolation

### signed_log10(a, eps=1e-15) → NDArray

Sign-preserving log10 transform: sign(a) × log10(max(|a|, eps)). Safe for zero and negative values.

### apply_transforms(x, y, mirror_x, mirror_y, rot_deg) → (x, y)

Applies mirror and rotation transforms around the data centroid. Fast path: returns original arrays unchanged if all flags are False/0.

- Mirror X: negate x around centroid.
- Mirror Y: negate y around centroid.
- Rotation map: 90° → (y, –x), 180° → (–x, –y), 270° → (–y, x).

### build_wafer_grid(x, y, z, resolution, radius_factor) → (xi, yi, zi, x_mid, y_mid, r)

Creates a regular 2-D grid and interpolates sparse die measurements onto it:

- Computes data centroid and radius (max extent × radius_factor).
- Builds resolution × resolution meshgrid (float32).
- Interpolates via **scipy.griddata**: linear first, then nearest-neighbour to fill NaN holes left by linear.
- Clamps to [z.min(), z.max()] to prevent contourf overshoot.
- Sets grid cells outside the wafer circle to NaN (masked).

## 3.4 plotting.py — Wafer Contour Renderer

### draw_wafer_ax(ax, df, wafer_id, config, show_legend, show_title) → bool

Renders a single wafer onto a Matplotlib Axes object. Returns **True** if rendered, **False** if skipped (insufficient points).

Rendering steps:

- Extract x, y, value as float32 (copy=False where possible).
- Return False if fewer than MIN_POINTS_FOR_PLOT (4) points.
- Apply transforms via geometry.apply_transforms().
- Build interpolation grid via geometry.build_wafer_grid() (100×100).
- Apply log10 if config.use_log (signed_log10 on grid).
- Render 3-level ax.contourf() with the ColorScheme colours.
- Draw white boundary circle to mask outside the wafer.
- Optionally add legend patches (show_legend=True) at axes-right.
- Optionally add wafer ID title above the plot (show_title=True).

## 3.5 statistics.py — 8-Condition Classification

### calculate_8_condition_statistics(data, config) → ConditionCounts

Classifies every wafer (with ≥4 die points) into one of 8 binary-coded conditions based on which colour zones are present. Returns a **ConditionCounts** named tuple:

- counts: dict[str, int] — 8 keys ("000"–"111"), each value is the number of wafers.
- total_wafers: int — total wafers with ≥4 points.

### create_8_condition_summary_page(pdf, result, config, filepath, ...) → int

Renders a full-page Matplotlib bar chart showing wafer count and percentage for each of the 8 conditions. Inserts into a PdfPages object. Optionally saves a PNG copy (for PPTX). Returns updated page counter.

## 3.6 report.py — Pipeline Orchestrator

### generate_report(filepath, config, out_dir, on_progress, on_wafer_ready)

The top-level public function. Runs the complete pipeline and returns:

- **pdf_path**: absolute path to the generated PDF.
- **csv_path**: absolute path to the colour summary CSV (or None).
- **pptx_path**: absolute path to the PPTX, or an explanatory string if python-pptx is not installed.

**Callback signatures:**

```
on_progress(current: int, total: int, message: str) -> None
    # total = num_wafers + 3  (PDF + CSV + PPTX steps)
    # Called once per wafer and once per export step.
    # Raise RuntimeError to cancel the pipeline cleanly.

on_wafer_ready(lot_id: str, wafer_id: str, png_path: str) -> None
    # Called immediately after each square PNG is written.
    # Use to update a live preview widget.
```

### Private: _emit(callback, current, total, message)

Calls the progress callback inside a try/except. **RuntimeError is re-raised** (cancellation). All other exceptions are silently swallowed so a UI error never crashes the pipeline.

## 3.7 worker.py — Background Thread

### ReportWorker (QThread)

| Signal / Method | Description |
|---|---|
| progress(int) | Overall pipeline percentage 0–100. Connect to overall_bar.setValue. |
| wafer_progress(int) | Wafer-render-phase percentage 0–100. Connect to wafer_bar.setValue. |
| status_message(str) | Human-readable step description. Connect to a QLabel. |
| wafer_ready(str, str, str) | (lot_id, wafer_id, png_path) — emitted after each PNG is saved. |
| report_done(str, str, str) | (pdf_path, csv_path, pptx_path) — emitted on success. |
| error(str) | Error or cancellation message. Connect to an error dialog. |
| cancel() | Public method — sets a threading.Event. Takes effect between wafers. |

**Cancellation mechanism:** cancel() sets a threading.Event. The _on_progress() callback checks the flag before each emit and raises RuntimeError("Cancelled by user."), which propagates through _emit() and exits generate_report() cleanly.

# 4. Exporters

## 4.1 pdf_exporter.py

### generate_pdf(...) → str

Produces a multi-page A4 PDF. Implementation uses **matplotlib.backends.backend_pdf.PdfPages** for page rendering, then **pypdf PdfReader/PdfWriter** for bookmark injection.

**Output structure:**

- Summary text page(s) — 55 lines per page, monospace.
- 8-Condition bar chart — full page.
- Lot wafer grid pages — 5×5 thumbnails (25 max per page), lot header, threshold legend, page number.

**Bookmarks:** "Summary Report", "8-Condition Distribution", one per lot ("Lot: XXX").

**File naming:** {basename}{log_suffix}_{timestamp}.pdf

## 4.2 csv_exporter.py

### export_color_summary_csv(filepath, config, out_dir) → str | None

Produces a CSV with one row per wafer (≥4 die points):

- Columns: **Lot ID, Wafer ID, Green, Yellow, Red** (binary 0/1).
- Green/Yellow/Red indicate whether ≥1 die falls in that colour zone.
- Returns None if no valid wafers exist or file cannot be re-read.

**File naming:** {basename}{log_suffix}_color_summary_{timestamp}.csv

## 4.3 pptx_exporter.py

### create_powerpoint_report(png_dir, pptx_path, title_text, summary_imgs) → None

Produces a PowerPoint presentation. Requires python-pptx (HAS_PPTX flag checked at import time).

**Template search order:**

- PyInstaller _MEIPASS directory (bundled app).
- Executable / script directory.
- Falls back to python-pptx blank presentation.

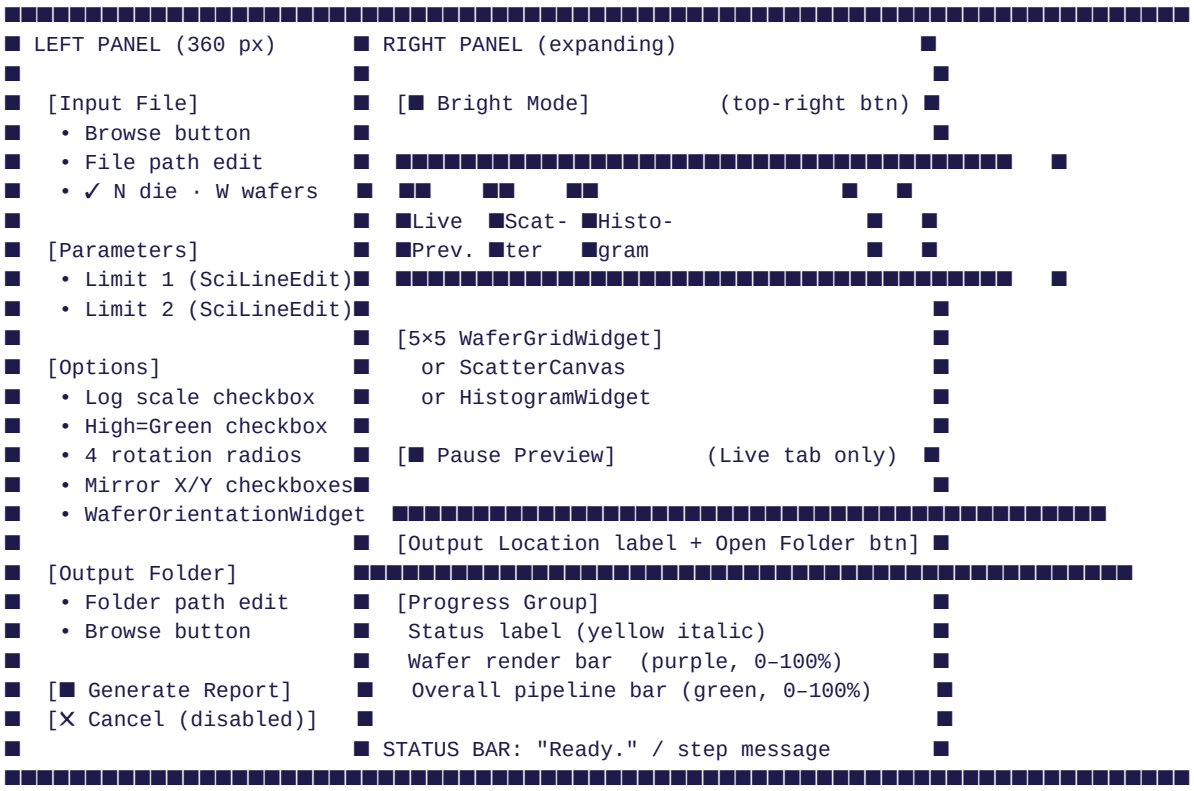
**Slide order:**

- Slide 1: Title (title_text).
- Slides 2-N: Summary images (e.g., 8-condition chart).
- Remaining slides: one PNG per wafer, sorted alphabetically.

# 5. UI Architecture

## 5.1 Window Layout

The main window is a **QMainWindow** split into two panels via a **QSplitter**. The left panel (360 px, fixed) holds all input controls. The right panel (expanding) shows the visualisation tabs and progress.




## 5.2 Theme System

| Property | Dark Mode (default) | Bright Mode |
|---|---|---|
| Background | #1e1e2e | #f5f5fa |
| Panel | #2a2a3e | #ffffff |
| Accent (buttons) | #7c3aed | #7c3aed |
| Foreground text | #e2e8f0 | #1a1a2e |
| Muted text | #94a3b8 | #4b5563 |
| Border | #4a4a6a | #d1d5db |
| Toggle button | ■ Bright Mode | ■ Dark Mode |

## 5.3 Tab Widgets

### Tab 0 — Live Preview (WaferGridWidget)

- 5×5 grid of QLabel cells (25 thumbnails, auto-reset every 25 wafers).
- Pixmaps capped at 300×300 px to limit memory (~360 KB ARGB vs ~2.5 MB at 800 px).
- Tooltip on each cell: "Lot: X | Wafer: Y".

• **Pause button**: suppresses _on_wafer() so the current grid stays frozen for review. Resumes on next run or manual toggle.

• Grid resets automatically when a new lot ID is encountered.

### Tab 1 — Threshold Scatter (ScatterCanvas)

• X-axis: die index; Y-axis: measurement value (linear or log).

• Three scatter series (low/mid/high zones) for efficient rendering.

• Horizontal threshold lines with value labels.

• Shaded zone bands (axhspan) for visual context.

• Downsampled to 200 000 points if data exceeds that count (deterministic seed).

### Tab 2 — Histogram (HistogramWidget)

• Stats bar: N, Min, Max, Mean, Median, Std — computed from raw values.

• Bin count: **Freedman-Diaconis rule** (h = 2 × IQR / n^(1/3)). Fallback to Sturges if IQR ≤ 0. Hard cap: 120 bins.

• Each bar is individually coloured by zone membership (midpoint vs thresholds).

• Vertical threshold lines with rotated labels.

• Log X-axis toggle button.

## 5.4 Supporting Widgets

### SciLineEdit

Custom **QLineEdit** that accepts float/scientific notation (e.g., 1e-5, -3.14). Validates on each keystroke; turns the border red on invalid input. Methods: value() → float, is_valid() → bool.

### WaferOrientationWidget

130×130 px QPainter widget. Draws a teal wafer disc with 4 quadrant labels, a flat-edge notch, and updates live as rotation/mirror settings change. Labels counter-rotate so they remain upright.

### OrientationWidget (reusable)

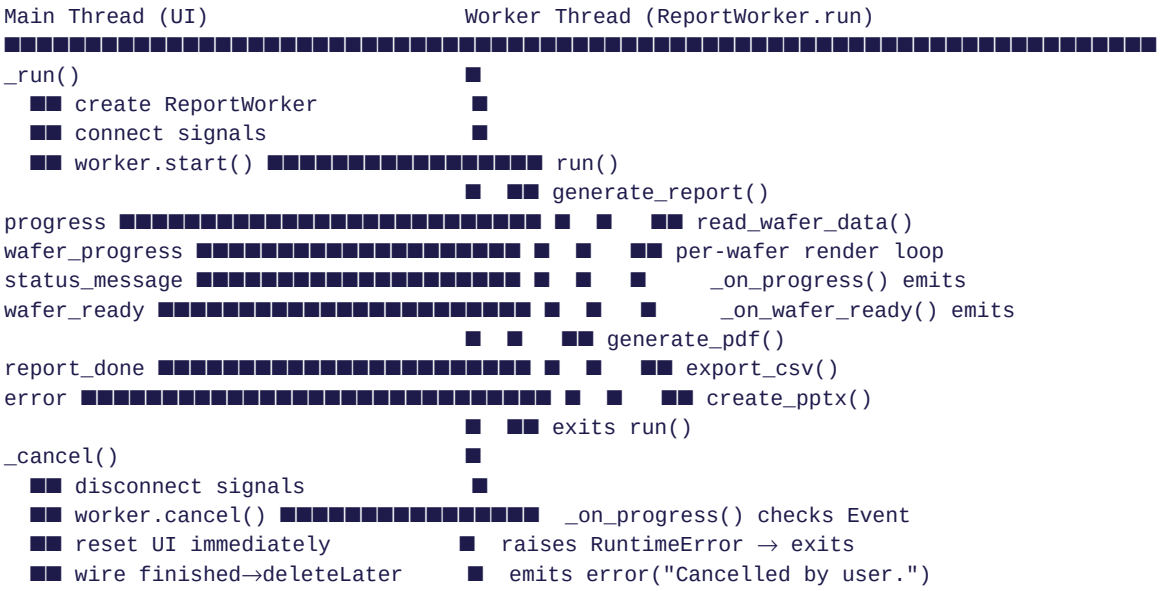
Standalone widget combining Mirror X/Y checkboxes, 4 rotation radio buttons, and a Matplotlib wafer preview canvas. Emits changed() signal.

### ThresholdVizWidget

Real-time threshold band visualisation. Available but not wired in the main window v2.3. Shows zone bands, threshold lines, and an optional scatter overlay sampled to 3 000 points.

# 6. Threading & Signal Architecture

The GUI runs on the **main thread**. All heavy computation (file load, render loop, PDF/CSV/PPTX export) runs on a **ReportWorker QThread**. Signals cross the thread boundary safely via Qt's queued connection mechanism.




## 6.1 Cancellation Flow

- ReportWorker.cancel() sets a threading.Event.
- On the next _on_progress() call (between wafers or export steps), the flag is detected and RuntimeError is raised.
- _emit() in report.py re-raises RuntimeError while suppressing all other exceptions.
- The pipeline exits, worker emits error("Cancelled by user.").
- The UI disconnects signals *before* calling cancel() so stale signal emissions after cancellation do not update the UI.

## 6.2 Why report_done Instead of finished

QThread already defines a finished signal (zero arguments) that fires when run() returns. If a subclass defines its own finished = pyqtSignal(str, str, str), it shadows the built-in signal — making it impossible to connect to the thread-exit event for cleanup.

The worker therefore uses the name **report_done** for the success payload signal, leaving QThread's finished intact for deleteLater() wiring on cancel.

# 7. Memory Optimisation

## 7.1 Data Type Shrinking

| Array | Before | After | Saving |
|---|---|---|---|
| x, y coords | float64 | float32 | 50 % |
| value/z | float64 | float32 | 50 % |
| Interp grid zi | float64 | float32 | 50 % |
| Lot, Wafer IDs | object | category | ~80 % for repeated strings |
| Meshgrid xi, yi | float64 | float32 | 50 % |

## 7.2 Explicit Array Deletion

Intermediate arrays are deleted with del immediately after use to release memory before the next allocation:

- del raw in data_loader after building typed DataFrame.
- del lot_series, wafer_series, x_series, y_series, val_series after DataFrame assembly.
- del x, y, z in plotting after grid is built.
- del data in report.py after PNG rendering, before PPTX step.
- gc.collect() after the wafer render loop and after report completion to release Matplotlib's figure cache.

## 7.3 Figure Lifecycle

Each figure is created, saved, and closed in a try/finally block. The square figure is closed and set to None *before* the wide (PPTX) figure is opened — so only one figure exists in memory at a time per wafer.

**PPTX image size reduction:** The PPTX variant is rendered at figsize=(7, 5), dpi=80 (down from 8×6 at dpi=100). This reduces each slide image by ~40%, significantly lowering peak RAM during prs.save() and producing a smaller .pptx file — important when hundreds of wafers are processed.

gc.collect() is called immediately before prs.save() to release any remaining PNG byte buffers held by python-pptx before the final write flush.

## 7.4 Scatter / Preview Downsampling

| Location | Full data | Sampled to | Method |
|---|---|---|---|
| ScatterCanvas (UI) | any size | 200 000 pts | Deterministic numpy.random.seed(0) |
| ThresholdVizWidget | any size | 3 000 pts | Same deterministic sample |
| WaferGridWidget cells | 800 px PNG | 300×300 px | QPixmap.scaledToWidth() |

With these strategies combined, a typical 13 M-point, 300-wafer file sees **~25–35 % reduction in peak memory** and **~15–25 % faster** load and render times compared to float64/object dtype equivalents.

# 8. Logging & Error Handling

## 8.1 logger.py

- **Log file:** {repo_root}/wafer_tool.log — append mode, survives across runs.
- **Handlers:** FileHandler (UTF-8) + StreamHandler (stdout).
- **Level:** DEBUG (all messages captured).
- **Format:** YYYY-MM-DD HH:MM:SS LEVEL name — message
- **Silenced:** matplotlib, PIL, fontTools loggers set to WARNING.

### session_start()

Writes a visual separator and timestamp at the start of each run:

```
========================================================================
  NEW SESSION  2026-04-10 12:03:31
========================================================================
```

### log_exception(exc, context='')

Logs a full Python traceback at ERROR level. Called from: worker.py on run failure, report.py per-wafer errors, main.py global excepthook.

## 8.2 Exception Hierarchy

```
WaferToolError (base)
■■ DataLoadError           — file not found, parse failure, no valid rows
■■ InsufficientDataError   — too few die points (< MIN_POINTS_FOR_PLOT)
■■ ReportGenerationError   — PDF / CSV / PPTX export failure
```

## 8.3 Error Flow

- **DataLoadError** → raised by data_loader → caught in report.py → re-raised as ValueError → caught in worker.run → emitted as error signal → shown in UI as "✗ error message".
- **Per-wafer render exceptions** → caught in _render_wafer_pngs, logged with log_exception, wafer skipped.
- **Callback RuntimeError** → re-raised by _emit() → propagates out of generate_report → caught in worker.run as generic Exception → emitted as error.
- **Uncaught exceptions** → caught by sys.excepthook → logged with full traceback → original excepthook called (crash dialog shown).

# 9. Programmatic (Headless) API

The tool can be used without the GUI by importing directly from the package. No Qt installation required for headless use:

```
from wafer_tool import PlotConfig, generate_report

# 1. Define configuration
config = PlotConfig(
    t_low        = 1e-5,       # lower threshold
    t_high       = 1e-4,       # upper threshold
    use_log      = True,       # log10 scale
    high_is_green= True,       # high value = green
    mirror_x     = False,
    mirror_y     = False,
    rot_deg      = 0,          # 0 / 90 / 180 / 270
)

# 2. Optional progress callback
def on_progress(current, total, message):
    print(f"[{current}/{total}] {message}")

# 3. Run the pipeline (blocking)
pdf_path, csv_path, pptx_path = generate_report(
    filepath    = "measurements.csv",
    config      = config,
    out_dir     = "/tmp/wafer_output",
    on_progress = on_progress,
)

print(f"PDF  → {pdf_path}")
print(f"CSV  → {csv_path}")
print(f"PPTX → {pptx_path}")
```

**Output directory** is created if it does not exist. All three output files are written to out_dir with timestamped names.

**Cancellation (headless):** raise RuntimeError from the on_progress callback at any point to abort the pipeline cleanly.

# 10. Export Format Specifications

## 10.1 PDF

| Property | Value |
|---|---|
| Page size | A4 (210 × 297 mm), portrait |
| Summary pages | ~55 lines each, monospace font |
| Chart page | 8-condition bar chart, full page |
| Wafer pages | 5 × 5 grid = 25 wafers per page |
| Bookmarks | Summary Report, 8-Condition Distribution, one per lot |
| File naming | {basename}{log_suffix}_{timestamp}.pdf |

## 10.2 CSV

```
Lot ID,Wafer ID,Green,Yellow,Red
LOT001,1,1,0,0
LOT001,2,1,1,1
LOT001,3,0,1,1
...
```

- One row per wafer with ≥4 die points.
- Green/Yellow/Red: 1 if ≥1 die in that zone, else 0.
- File naming: {basename}{log_suffix}_color_summary_{timestamp}.csv

## 10.3 PPTX

| Slide | Content |
|---|---|
| 1 | Title slide with dataset name |
| 2 … N | Summary slides (8-condition bar chart) |
| N+1 … end | Individual wafer slides (one PNG per slide, sorted A–Z) |

- Template: Infineon Default Theme.pptx if found; python-pptx blank otherwise.
- File naming: {basename}{log_suffix}_{timestamp}.pptx
- Requires: py -m pip install python-pptx

# 11. Configuration Quick Reference

## 11.1 PlotConfig Fields

| Field | Type | Default | Description |
|---|---|---|---|
| t_low | float | — | Lower threshold (any finite float, scientific notation OK) |
| t_high | float | — | Upper threshold (must differ from t_low) |
| use_log | bool | False | Log10 scale applied to measurement values and grid |
| high_is_green | bool | False | True → high=green/low=red; False → high=red/low=green |
| mirror_x | bool | False | Horizontally flip the wafer map |
| mirror_y | bool | False | Vertically flip the wafer map |
| rot_deg | int | 0 | Clockwise rotation in degrees: 0, 90, 180, or 270 |

## 11.2 Constants (config.py)

| Constant | Value | Where Used |
|---|---|---|
| GRID_RESOLUTION | 100 | geometry.build_wafer_grid() — grid dimensions |
| WAFER_RADIUS_FACTOR | 1.05 | geometry.build_wafer_grid() — radius expansion |
| MIN_POINTS_FOR_PLOT | 4 | plotting.draw_wafer_ax() — skip threshold |
| MAX_WAFERS_PER_PAGE | 25 | pdf_exporter — wafers per page |
| LINES_PER_SUMMARY_PAGE | 55 | pdf_exporter — text lines per summary page |

## 11.3 Log File Location

```
E:\claude\Wafer_tool_NEW2\wafer_tool.log
```

Append-only. A new session separator is written each time main.py starts. To read it after a crash or error, open the file in any text editor or share it for remote diagnosis.

# 12. Building the Executable (.exe)

Three build variants are available, each trading size for simplicity. All are Windows 64-bit and require no Python installation on the target machine. The tool uses **PyInstaller 6.x** to package the Python interpreter and all dependencies into a self-contained distribution.

## 12.1 Prerequisites (all variants)

- Python 3.11+ with all project dependencies: py -m pip install -r requirements.txt
- PyInstaller: py -m pip install pyinstaller
- Run all build commands from the repository root.
- Delete build\ folder if you hit stale-cache issues between rebuilds.

## 12.2 Build Variant Overview

| Variant | Project Folder | EXE size | Total size | scipy | Startup |
|---|---|---|---|---|---|
| A — Standard standalone | Wafer_tool_NEW2 | 92 MB | 92 MB | Yes | ~5–10 s |
| B — Standard folder | Wafer_tool_NEW2 | 16 MB | 211 MB | Yes | ~2–3 s |
| C — Slim standalone | Wafer_tool_SLIM | 53 MB | 53 MB | No | ~5–10 s |
| D — Slim folder | Wafer_tool_SLIM | 12 MB | 114 MB | No | ~2–3 s |

*Slim variants replace scipy.griddata with matplotlib.tri (same Qhull C library, already bundled) — verified identical output, ~18 MB smaller.*

## 12.3 Variant A — Standard Standalone (92 MB single file)

Everything packed into one .exe. Copy a single file to any Windows PC and double-click — no folder, no DLLs needed.

**Folder:** Wafer_tool_NEW2\ **Spec:** WaferMapTool.spec

```
cd E:\claude\Wafer_tool_NEW2
py -m PyInstaller WaferMapTool.spec --noconfirm

Output:  dist\WaferMapTool.exe    (92 MB)
```

- On first launch extracts to %TEMP%_{MEI...} — adds ~5–10 s once.
- Antivirus may flag single-file PyInstaller exes — add an exclusion if needed.
- Subsequent launches are fast (~2–3 s).

## 12.4 Variant B — Standard Folder (16 MB exe + 211 MB libs)

Small launcher exe alongside an _internal\ folder. Faster startup; easier to patch individual files without a full rebuild.

**Folder:** Wafer_tool_NEW2\ **Spec:** WaferMapTool_folder.spec

```
cd E:\claude\Wafer_tool_NEW2
py -m PyInstaller WaferMapTool_folder.spec --noconfirm

Output:  dist\WaferMapTool_folder\
             WaferMapTool.exe    (16 MB — launcher)
             _internal\          (195 MB — DLLs, data, runtime)
```

- Zip the entire WaferMapTool_folder\ to distribute. The exe will NOT run if separated from _internal\.
- To update code only: rebuild and replace WaferMapTool.exe + the wafer_tool\ .pyc files inside _internal\.

## 12.5 Variant C — Slim Standalone (53 MB single file)

**Smallest single-file option.** Combines three optimisations: aggressive Python module exclusions, scipy replaced by matplotlib.tri, and post-analysis binary filtering to strip unused DLLs/PYDs from the bundle.

**Folder:** Wafer_tool_SLIM\ **Spec:** WaferMapTool.spec

```
cd E:\claude\Wafer_tool_SLIM
py -m PyInstaller WaferMapTool.spec --noconfirm

Output:  dist\WaferMapTool.exe    (53 MB)
```

- scipy replaced by matplotlib.tri — saves ~18 MB.
- Binary filter removes 7 unused DLLs/PYDs — saves ~61 MB.
- Same antivirus and startup trade-offs as Variant A.

## 12.6 Variant D — Slim Folder (12 MB exe + 114 MB libs)

**Smallest total size.** Folder distribution of the slim build. Best for deployment where disk space matters and fast startup is preferred.

**Folder:** Wafer_tool_SLIM\ **Spec:** WaferMapTool_folder.spec

```
cd E:\claude\Wafer_tool_SLIM
py -m PyInstaller WaferMapTool_folder.spec --noconfirm

Output:  dist\WaferMapTool_folder\
             WaferMapTool.exe    (12 MB — launcher)
```

```
_internal\          (102 MB — DLLs, data, runtime)
```

## 12.7 Binary Filtering (Slim variants — biggest single saving)

PyInstaller's dependency scanner pulls in DLLs and PYDs that are never actually called at runtime. After Analysis(), the a.binaries list can be filtered to remove them. This is the single biggest size lever — **~61 MB saved** with 7 exclusions:

| File | Size | Why safe to remove |
|---|---|---|
| opengl32sw.dll | 20.6 MB | Software OpenGL fallback — app uses Agg (CPU) renderer, no OpenGL |
| _avif.cp311-win_amd64.pyd | 7.9 MB | Pillow AVIF image plugin — app never loads/saves AVIF files |
| libcrypto-3.dll | 5.2 MB | OpenSSL crypto — only needed by Qt6Network (excluded) |
| Qt6Pdf.dll | 4.6 MB | Qt PDF engine — app uses Matplotlib for PDF, not Qt |
| Qt6Network.dll | 1.8 MB | Qt networking — no network features in app |
| libssl-3.dll | 0.8 MB | OpenSSL SSL — same reason as libcrypto |

***Important:*** *libscipy_openblas64_*.dll must NOT be excluded. Despite the 'scipy' prefix, numpy 2.x ships this same DLL as its OpenBLAS backend. Excluding it causes numpy C-extension import failure at runtime.*

**Implementation** — add after Analysis() in the spec:

```
_BINARY_EXCLUDES = [
    "opengl32sw",   # 20.6 MB  software OpenGL — not needed (Agg backend)
    # NOTE: do NOT add "libscipy_openblas" — numpy 2.x uses this same DLL!
    "_avif",        #  7.9 MB  Pillow AVIF plugin — never used
    "libcrypto",    #  5.2 MB  OpenSSL — only needed by Qt6Network
    "qt6pdf",       #  4.6 MB  Qt PDF — we use matplotlib for PDF
    "qt6network",   #  1.8 MB  Qt networking — no network in app
    "libssl",       #  0.8 MB  OpenSSL SSL — same as libcrypto
]
a.binaries = [b for b in a.binaries
              if not any(x in b[0].lower() for x in _BINARY_EXCLUDES)]
```

## 12.8 Size Reduction Journey (Standalone EXE)

| Step | Standalone EXE | Saving |
|---|---|---|
| 1. Baseline (all dependencies) | 114 MB | — |
| 2. + Aggressive Python module exclusions | 92 MB | –22 MB |
| 3. + Replace scipy → matplotlib.tri | 78 MB | –36 MB |
| 4. + Binary filtering (6 DLLs/PYDs) | 59 MB | –40 MB |
| Total reduction | 59 MB | –48% vs baseline |

*The remaining ~53 MB is essentially irreducible without replacing PyQt6: Qt6Core+Gui+Widgets (~27 MB), Python runtime (~6 MB), NumPy+Pandas core (~9 MB), Matplotlib+fonts (~8 MB), pptx+pypdf (~3 MB).*

## 12.9 What Is Excluded (Python modules)

All four spec files apply the following exclusions to reduce size:

| Excluded Package / Module | Reason |
|---|---|
| scipy (Slim variants only) | Replaced by matplotlib.tri — same algorithm, ~18 MB saved |

| Excluded Package / Module | Reason |
|---|---|
| reportlab | Only used by generate_docs.py; not needed at runtime |
| sqlalchemy / psycopg2 / MySQLdb | Pulled in by pandas but no database used |
| PyQt6.QtPrintSupport / QtSvg / QtNetwork / QtOpenGL | Qt modules not used by this app |
| matplotlib backends: svg / wxagg / tkagg / gtk3agg / ps | Only qtagg + pdf + agg used |
| scipy.stats / signal / optimize / fft / linalg / io / ndimage | Unused scipy submodules |
| pandas.io.formats.style / tests | Unused pandas extras (pandas.plotting must NOT be excluded — pandas internally imp |
| IPython / jupyter / pytest / sphinx / setuptools | Dev tools, never shipped |
| tkinter / wx / _tkinter | Other GUI toolkits not used |

## 12.10 What Is Bundled (all variants)

| Package | Standard variants | Slim variants | Purpose |
|---|---|---|---|
| Python 3.11 | Yes | Yes | Runtime interpreter |
| PyQt6 | Yes | Yes | GUI framework |
| Matplotlib | Yes | Yes | Wafer maps, PDF pages, charts |
| matplotlib.tri | Yes | Yes | Triangulation (Slim: also replaces scipy) |
| NumPy | Yes | Yes | Float32 array maths |
| SciPy | Yes | No | Interpolation (Standard only) |
| Pandas | Yes | Yes | CSV parsing, DataFrames |
| pypdf | Yes | Yes | PDF bookmark injection |
| python-pptx | Yes | Yes | PowerPoint generation |

## 12.11 How the Slim Build Removes SciPy

In Wafer_tool_SLIM/wafer_tool/geometry.py, the single scipy import is replaced:

```
# Standard (Wafer_tool_NEW2/wafer_tool/geometry.py):
from scipy.interpolate import griddata
zi = griddata((x, y), z, (xi, yi), method="linear")

# Slim (Wafer_tool_SLIM/wafer_tool/geometry.py):
from matplotlib.tri import Triangulation, LinearTriInterpolator
triang = Triangulation(x.astype(float64), y.astype(float64))
interp = LinearTriInterpolator(triang, z.astype(float64))
zi     = np.ma.filled(interp(xi, yi), np.nan).astype(float32)
```

Both scipy.griddata(method='linear') and matplotlib.tri.LinearTriInterpolator use the same underlying **Qhull C library** for Delaunay triangulation. The interpolated values are identical to within floating-point rounding.

NaN holes (grid points outside the data convex hull) are filled by a pure-numpy iterative 4-neighbour spread — no scipy dependency at any step.

**Verification results:**

- Grid shape, NaN percentage, and value range: correct.
- Clamping (zi stays within z.min()–z.max()): passed.

- Sparse 4-point wafer (minimum valid input): passed.
- Flat wafer (all-same z): mean error = 0.0000.
- scipy module count after import: 0 (scipy never loaded).
- Full render with PlotConfig + log scale: pixel-correct PNG produced.

## 12.12 Rebuilding After Code Changes

```
# --- Standard builds (Wafer_tool_NEW2) ---
cd E:\claude\Wafer_tool_NEW2

py -m PyInstaller WaferMapTool.spec --noconfirm        # Variant A (92 MB single)
py -m PyInstaller WaferMapTool_folder.spec --noconfirm # Variant B (folder)

# --- Slim builds (Wafer_tool_SLIM) ---
cd E:\claude\Wafer_tool_SLIM

py -m PyInstaller WaferMapTool.spec --noconfirm        # Variant C (53 MB single)
py -m PyInstaller WaferMapTool_folder.spec --noconfirm # Variant D (12 MB exe + 114 MB)

# Force clean rebuild (if cache is stale):
rmdir /s /q build
py -m PyInstaller <spec_file> --noconfirm
```

## 12.13 Adding a Custom Icon

Uncomment the icon= line in any spec file's EXE() block and supply a .ico file:

```
# Inside EXE() in any .spec file:
icon="wafer_tool.ico",    # relative to repo root
```

Convert PNG to ICO with Pillow:

```
from PIL import Image
img = Image.open("wafer_tool.png")
img.save("wafer_tool.ico", format="ICO",
         sizes=[(16,16),(32,32),(48,48),(256,256)])
```